Experiment 8: Buck Converter

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Aim

To design and test a buck converter using the TL494 PWM module.

Specifications

Input: 15–25 V (20 V nominal), output: 10 V, switching frequency: 25 kHz, rated output power: 100 W. The current and voltage ripples are taken as 0.2 A and 0.1 V peak-to-peak.

Theory and calculations

A buck converter gives a lower DC output voltage by controlling the switch duty ratio. The inductor and capacitor smooth the switched waveform. For ideal operation,

$$V_o = DV_{in}, \qquad D = \frac{10}{20} = 0.5 = 50\%.$$

The switching period is $T = 1/(25 \times 10^3) = 40\,\mu$s, so the ON and OFF times are both $20\,\mu$s at the nominal input.

The rated output current and load resistance are

$$I_o = \frac{P_o}{V_o} = \frac{100}{10} = 10\,\text{A}, \qquad R = \frac{V_o^2}{P_o} = \frac{10^2}{100} = 1\,\Omega.$$

The required inductance at 20 V is

$$L = \frac{(V_{in} - V_o)D}{f_s \Delta I_L} = \frac{(20-10)\times 0.5}{25000 \times 0.2} = \boxed{1\,\text{mH}}.$$

Neglecting capacitor ESR, the minimum output capacitance is

$$C = \frac{\Delta I_L}{8 f_s \Delta V_o} = \frac{0.2}{8 \times 25000 \times 0.1} = \boxed{10\,\mu\text{F}}.$$

For the practical inductor, $L = 0.95$ mH, the expected current ripple at 20 V is

$$\Delta I_L = \frac{(20-10)\times 0.5}{25000 \times 0.95 \times 10^{-3}} \approx \boxed{0.211\,\text{A}}.$$

These calculations are for the nominal input. To meet the 0.2 A current-ripple limit throughout 15–25 V, the required inductance is at least 1.2 mH.

Procedure and observations

The converter was operated using a TL494 module, an IRF540N MOSFET, a fast recovery diode and the 0.95 mH inductor. The waveforms were recorded on the oscilloscope. The probable gate waveform shows 26.04 kHz and 53.71% duty. The likely output-voltage trace has a mean of 10.12 V. Switching spikes are visible in the pulse waveform. The probe points are not labelled in the photographs, so these signal identifications are inferred.

Result

The recorded output is consistent with the 10 V target. The practical inductor gives a calculated ripple close to the specified 0.2 A at nominal input. Current ripple and 100 W output operation were not verified from the available photographs.

Recorded waveforms

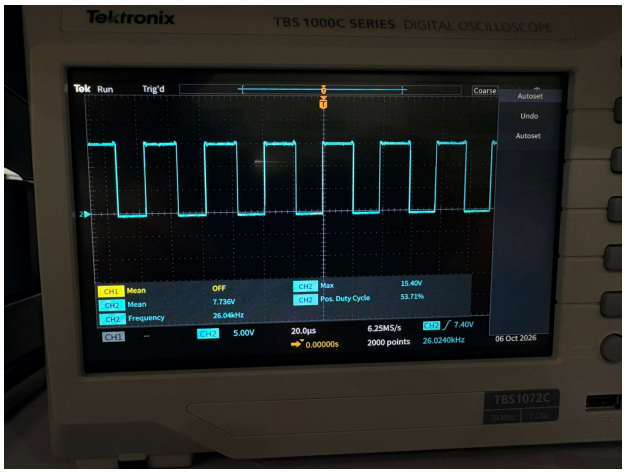


Figure 1: Probable gate pulses: 26.04 kHz, 53.71% duty.

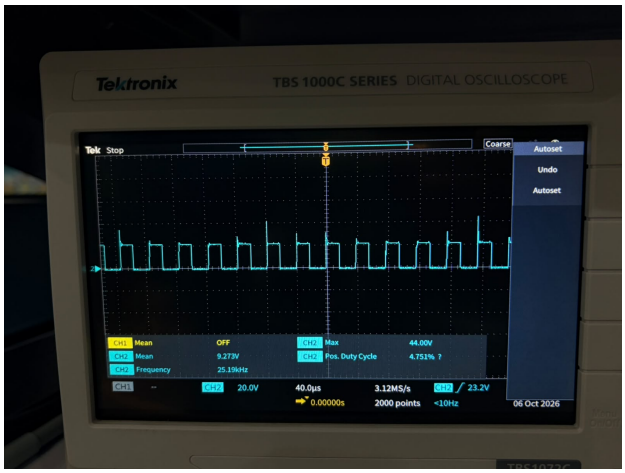


Figure 2: Switching-voltage waveform with spikes. Exact probe point is unknown.

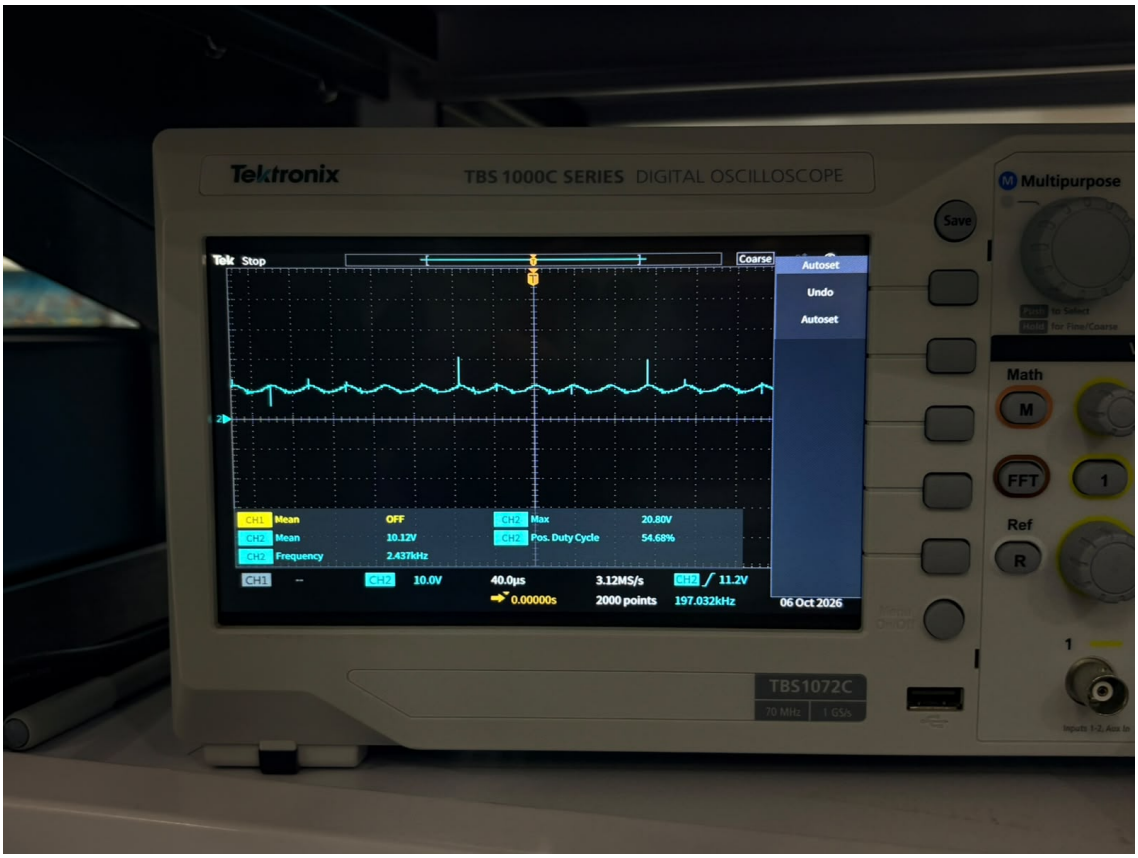


Figure 3: Probable output voltage. The displayed mean is 10.12 V.

The photographs show PWM switching and an approximately DC output. Some automatic readings on the noisy traces are unreliable; the output ripple cannot be checked accurately at the displayed scale.

Reference: DPC 2026 Experiment 8 laboratory sheet.